

Question 1 - Set Creation

Create a set containing 10 programming languages.

Print:

- Complete set
 - Total elements
-

Question 2 - Remove Duplicates

Given:

```
numbers = [1,2,2,3,4,5,5,6,6,7]
```

Convert it into a set and print unique values.

Question 3 - Union

Create two sets of student names.

Print:

- Union
-

Question 4 - Intersection

Create two sets representing students enrolled in Python and Java.

Print students enrolled in both courses.

Question 5 - Difference

Using the above sets, print:

- Students only in Python
 - Students only in Java
-

Question 6 - Symmetric Difference

Print students enrolled in exactly one course.

Question 7 - Dictionary Basics

Create a dictionary containing details of a student:

- Name
- Age
- City
- Course

Print all key-value pairs.

Question 8 - Dictionary Update

Given:

```
student = {  
    "name": "Rahul",  
    "age": 20  
}
```

- Add city
- Add marks
- Update age

Print the dictionary.

Question 9 - Dictionary Traversal

Print:

- All keys
 - All values
 - Key-value pairs
-

Question 10 - Student Result

Create a dictionary of student marks.

Print:

- Highest scorer
- Lowest scorer

Question 11 - Word Frequency

Accept a sentence from the user. Create a dictionary storing frequency of every word.

Question 12 - Character Frequency

Accept a string and count frequency of each character.

Question 13 - Merge Dictionaries

Create two dictionaries and merge them into one.

Question 14 - Inventory Management

Create a dictionary representing a shop inventory.

Example:

```
Laptop : 15  
Mouse : 40  
Keyboard : 25
```

Update stock after selling items.

Question 15 - Student Database

Create a dictionary where:

```
Roll Number → Student Name
```

Search a student using roll number.

Question 16 - Nested Dictionary

Create a nested dictionary containing details of five students.

Each student should contain:

- Name
- Age
- Marks

Print complete information.

Question 17 - Dictionary Sorting

Sort a dictionary:

- By keys
 - By values
-

Question 18 - Mini Project

Create a Contact Management System using dictionary.

Features:

- Add contact
- Delete contact
- Update contact
- Search contact
- Display all contacts